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Effect of generator De-rating operations on torsional vibrations and torsional interactions with a HVDC transmission system

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ABSTRACT

In this paper, the influence of generator de-rating operations on the SSTI (Sub-Synchronous Torsional Interactions) between torsional modes and controls of a LCC—HVDC (Line-Commutated Converter High-Voltage Direct Current) transmission system is studied. The modified CIGRE HVDC benchmark system is adopted for the studies. The simulation analyses (including modal analysis and transient simulations) are made by using the DIgSILENT PowerFactory software. It is found that the turbine shafts torsional responses will be significantly affected by the generator de-rating operations in a negative way. Following the decrease in generation power, some of the turbine shafts torsional modes might become unstable. Furthermore, the SSTI between torsional modes and HVDC controls is also significantly affected by the generator de-rating operations. It may aggravate the negative effect of SSTI, changing the effect from non to negative or changing the effect from positive to negative. Moreover, the de-rating operations of a generator unit could even exacerbate the SSTI between the torsional modes of the neighboring generator unit and the HVDC controls. Also, the influence is negative.

1. Introduction

Recently, the SSTI (Sub-Synchronous Torsional Interaction) has become a hot topic of researches. The most common example of SSTI is found at the rectifier side of a HVDC (High-Voltage Direct Current) transmission system. When connecting the rectifier side of a HVDC transmission link to an AC network with a turbine generator, the rectifier contributes with negative damping in the sub-synchronous frequency range. The first experience of HVDC interaction with turbine-generator torsional vibration occurred at Square Butte Electric Coop Project in North Dacota in 1977. It showed that the relatively high gain power modulation control did indeed de-stabilize the first torsional vibration mode. After that, a lot of investigations were carried out to ensure a net positive damping contribution to torsional modes of vibration for all turbine-generator units in the vicinity of a HVDC system. In [1], it used the full-scale model of a practical modular multilevel converter based VSC—HVDC (Voltage Source Converter HVDC) and the multi-mass shaft model of a turbo-generator to analyze the characteristics of SSTI. And the modal signal injection test method was proposed to identify the damping of torsional modes and the impact from decisive factors such as the loading levels, control strategies of converters, and the coupling strength between VSC—HVDC and ac networks. In [2], SSTI with VSC—HVDC affected by voltage feed forward compensations in the AC current controllers was investigated. A general output impedance model of VSC—HVDC concentrated on AC current controllers was developed first. Then the influence of compensation factor and filter bandwidth on the bandwidth of the converter and the electrical damping was studied. Finally, a SSTI suppression scheme based on the former analysis was given. In [3], SSTI phenomenon for hydroelectric turbo-generator units with small generator-to-turbine inertia ratios interconnected to an HVDC system was investigated. The sensitivity of different system parameters, such as generating capacity and loading, DC power flow, AC network strength and current controller gains, towards electrical damping contribution were examined. In addition to the HVDC system, others fast power regulation equipment like power system stabilizer, as well as FACTS (Flexible AC Transmission System) devices like SVC (Static VAR Compensator), TCSC (Thyristor-Controlled Series Capacitor), static compensator, unified power flow controller… etc. were also found to possibly cause the SSTI problems. In [4,5], it showed that the installation of a SVC in Chester, Maine might have a negative influence on the stability of torsional modes of vibration of the nearby turbine generators. In [6], a general approach based on interconnected dynamical system theory and linear analysis techniques was proposed for the analysis and characterization of SSTI with FACTS devices. The influence of control strategy and mode of operation of FACTS controllers on the damping of torsional modes was discussed. In [7], it

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introduced, examined, and evaluated the industrial experiences and effectiveness of a TCSC replica controller installed in Korea. In [8], it showed that frequent oscillation occurs at the wind farms of Santanghu District, which were connected to $a \pm 800$ kV HVDC line. After analyzing the oscillation frequency and its characteristics, it was concluded that the oscillation occurs because of SSTI caused by the interaction between the wind turbine generators and the HVDC link. To suppress the SSTI phenomenon, an additional SSTI stabilizer was proposed and installed.

Recently, wind farms have expanded extensively. It also gives rise to the SSTI problems [9,10]. When a large scale wind turbine generator is incorporated into a system with the FACTS devices, there is potentially the SSTI between wind turbine mechanism and FACTS controllers [11]. Besides, the power controller of wind turbine generators itself actually is a FACTS device. So, even though there are not any other FACTS devices in the system, the power controller of wind turbine generators could possibly interact with the turbine-and-generator mechanism of the traditional generator units nearby [12]. On the contrary, there are investigations made in the opposite direction. A supplementary controller was added to the wind power controller to increase the damping of torsional modes, thus improving the torsional responses of a generator [13].

According to the surveys above, the SSTI might potentially occur between the wind turbine mechanism and the FACTS controller, as well as between the wind power controller and the steam turbine mechanism when incorporating wind farms into power systems. Both directly involve the wind power generator units themselves. It was ignored, however, the indirect impact of integrating a wind farm on the SSTI. In this paper, it is aimed to study such an effect. A large scale wind farm is incorporated into a HVDC system with limited transmission capacity. The HVDC transmission power is controlled not to exceed the limitation. The steam turbine generator unit thus needs to operate with de-rated power to counterbalance the wind farm power. Firstly, the effect of such a de-rating operation on torsional modes is studied. It is found that it affects the damping ratio of torsional modes in a negative way, leading to greater vibrations and even dynamics instability in case of faults. Then, the influence of de-rating operations on the effect of SSTI with HVDC controls is studied. It is found that the influence is also negative. It could aggravate the negative effect of SSTI with HVDC controls, changing the effect from non to negative or even changing the effect from positive to negative. Finally, the influence on the SSTI of a neighboring unit with HVDC controls is studied. It is found that the de-rating operation of a unit could exacerbate the effect of SSTI between the HVDC controls and the torsional modes of the neighboring unit, and the influence is negative. Furthermore, the effect of generator de-rating operations and the effect of SSTI with HVDC controls are found to be cumulative, hence attention should be paid to the double action of these two effects. However, the effect of SSTI with HVDC controls is usually far smaller than the effect of generator de-rating operations. Therefore, the stability of torsional responses is, in fact, dominated by the effect of generator de-rating operations.

2. System studied

2.1. System descriptions

The wind-thermal bundling system studied is shown in Fig. 1, which is modified from the CIGRE HVDC benchmark system. The thermal power plant and the wind farm connect at the 345 kV bus, supplying power to the HVDC transmission system. On the other side of HVDC system is a 230 kV network. The HVDC transmission system is a monopolar 500 kV, 1000 MW HVDC link with 12-pulse thyristor converters (two 6 pulse converters in series) on both the rectifier and inverter sides. The 500 kV DC line has a length of 500 km and is rated at 2 kA. The capacity is limited to 1000 MW. The thermal power plant consists of two steam turbine generator units G1 and G2. Each generator unit is with rated power of 892 MVA. The wind farm consists of 140 wind turbines, which are equipped with doubly-fed induction generator (DFIG). Each wind turbine has a nominal active power of 5 MW. The 140 wind turbines are aggregated into seven DFIG models when simulation studying. Each model represents 20 parallel machines, and is 100 MW of active power.

2.2. System model and analysis method

The DIgSILENT PowerFactory software is adopted for the analysis and simulations. Built-in models are employed to implement the system studied.

2.2.1. Modeling of HVDC system

The DIgSILENT PowerFactory has a built-in template of LCC—HVDC transmission system. Some modifications have been made to fit the requirement of CIGRE HVDC bench mark system.

2.2.1.1. AC system. The converter transformers are implemented by using the three phase two winding transformer, one with grounded Y-Y connection and the other with grounded Y-△ connection. The AC filters, including high-pass filters, low-pass filters, and shunt capacitors are also implemented by the standard elements. Those models are common and will be not repeated here.

2.2.1.2. DC transmission line. In Fig. 2, it shows the DC transmission line

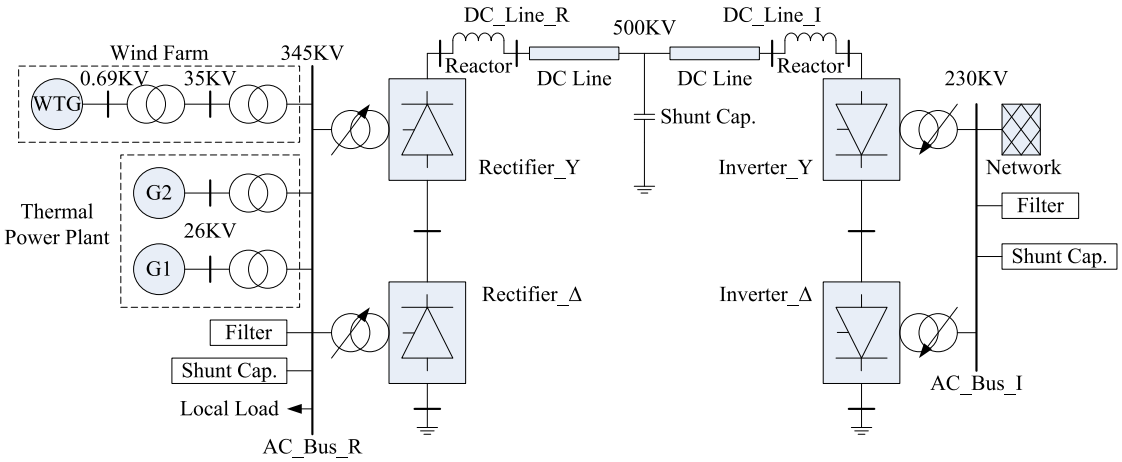


Fig. 1. System studied.

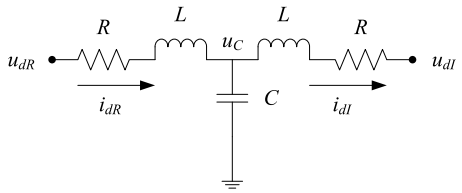


Fig. 2. Model of DC transmission line.

model adopted, which is an equivalent T network integrating with the converters. The dynamic equations can be obtained as the followings.

$$C\frac{du_C}{dt} = i_{dR} - i_{dI} \tag{1}$$

$$L\frac{di_{dR}}{dt} = u_{dR} - u_C - Ri_{dR} \tag{2}$$

$$L\frac{di_{dI}}{dt} = u_C - u_{dI} - Ri_{dI} \tag{3}$$

$$u_{dR} = \frac{3\sqrt{2}}{\pi}u_{acR}\cos(\alpha_R) - 3\frac{w_R L_{acR}}{\pi}i_{dR} \tag{4}$$

$$u_{dI} = -\frac{3\sqrt{2}}{\pi}u_{acI}\cos(\alpha_I) + 3\frac{w_I L_{acI}}{\pi}i_{dI} \tag{5}$$

u_{dR},u_{dI}: DC terminal voltages of the rectifier and the inverter

i_{dR},i_{dI}: DC currents of the rectifier and the inverter

u_C: DC link voltage

L_{acR},L_{acI}: equivalent inductances for the rectifier and the inverter AC terminals

u_{acR},u_{acI}: line-to-line voltages at the rectifier and the inverter AC terminals.

w_R,w_I: frequencies of the rectifier and the inverter AC systems

2.2.1.3. Rectifier control. The rectifier control is composed of constant-current control and minimum trigger angle limiter. In Fig. 3, it shows the control scheme. Under normal conditions, the controller regulates the DC current to the current reference i_{dR_ref}. In the event of a severe drop in the DC voltage the current reference is reduced to i_{dR_VDCOL} through the VDCOL (Voltage Dependent Current-Order Limiter). Traditional PI type of current controller is adopted, with proportional gain K_p and integral time constant T_i. The dynamic equations are as the followings.

$$\frac{K_{fir}}{T_{fir}}i_{dR} = \frac{1}{T_{fir}}i_{dR_meas} + \frac{di_{dR_meas}}{dt} \tag{6}$$

$$\alpha_R = \pi - \beta_R = \pi - \left[K_p\left(i_{dRref} - i_{dR_meas}\right) + \frac{1}{T_i}\int\left(i_{dRref} - i_{dR_meas}\right)dt\right] \tag{7}$$

K_{fir},T_{fir}: gain and time constant of filter

K_p,T_i: proportional gain and integral time constant of controller

i_{dR_meas},i_{dR_ref},i_{dR_VDCOL}: measured dc current of rectifier, the reference dc current and the reference dc current coming from VDCOL

α_R: firing angle of rectifier

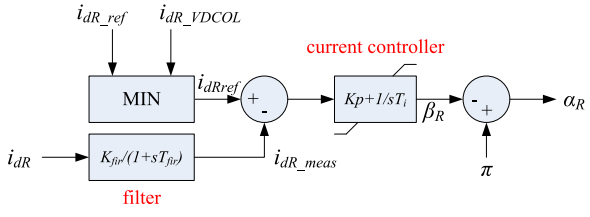


Fig. 3. Rectifier control model.

2.2.1.4. Inverter control. The inverter control is composed of constant-extinction angle control, constant-current control and current error control, In Fig. 4, it shows the control scheme. Under normal conditions, the controller regulates the extinction angle to γ_{min}. In the event of a severe reduction in DC voltage, the controller can switch to current-control mode. In this case the inverter regulates the DC current. Current error control is an auxiliary control regulator between constant-current and constant-extinction angle controls. When the difference between the current measured value and the current setting value on the inverter is too large, the current error controller will output a larger value and add it to the constant-extinction angle control, thereby γ will increase rapidly. Traditional PI type of extinction angle controller is adopted, with proportional gain K_{p_gamma} and integral time constant T_{i_gamma}. The dynamic equations are as the followings. constant-extinction angle control mode:

$$\frac{K_{f\gamma}}{T_{f\gamma}}\gamma = \frac{1}{T_{f\gamma}}\gamma_{meas} + \frac{d\gamma_{meas}}{dt} \tag{8}$$

$$\begin{aligned}\alpha_I &= \pi - \beta_I \\ &= \pi \\ &\quad - \left[K_{p_gamma}(\gamma_{min} + \gamma_{CEC} - \gamma_{meas})\gamma_{err} + \frac{1}{T_{i_gamma}}\int(\gamma_{min} + \gamma_{CEC} - \gamma_{meas})dt\right]\end{aligned} \tag{9}$$

$K_{f\gamma}$,$T_{f\gamma}$: gain and time constant of filter

K_{p_gamma},T_{i_gamma}: proportional gain and integral time constant of extinction angle controller

γ,γ_{min},γ_{meas}: extinction angle of inverter, minimal extinction angle and measured extinction angle

γ_{CEC}: output signal of the current error controller

α_I: firing angle of inverter constant-current control mode:

$$\frac{K_{fii}}{T_{fii}}i_{dI} = \frac{1}{T_{fii}}i_{dI_meas} + \frac{di_{dI_meas}}{dt} \tag{10}$$

$$\begin{aligned}\alpha_I &= \pi - \beta_I \\ &= \pi - \left[K_{p_curr}\left(i_{dI_ref} - i_{dI_meas} - i_m\right) + \frac{1}{T_{i_curr}}\int\left(i_{dI_ref} - i_{dI_meas} - i_m\right)dt\right]\end{aligned} \tag{11}$$

K_{fii},T_{fii}: gain and time constant of filter

K_{p_curr},T_{i_curr}: proportional gain and integral time constant of current controller

i_{dI_meas},i_{dI_ref}: measured dc current of inverter and the reference dc current

i_m: current margin at the inverter side

2.2.2. Modeling of DFIG unit

2.2.2.1. Induction generator. The DFIG model adopted is a model which integrates the induction machine and rotor-side converter. During time domain simulations the converter is controlled through the pulse width modulation factors. The grid-side converter is neglected, and the DC link voltage is assumed to be constant. The voltage equations are as followings.

$$\underline{u}_s = \underline{R}_s\underline{i}_s + j\frac{w_{syn}}{w_n}\underline{\psi}_s + \frac{1}{w_n}\frac{d\underline{\psi}_s}{dt} \tag{12}$$

$$\underline{u}_r e^{-j\left(w_{syn}-w_r\right)t} = \underline{R}_r\underline{i}_r + j\frac{w_{syn}-w_r}{w_n}\underline{\psi}_r + \frac{1}{w_n}\frac{d\underline{\psi}_r}{dt} \tag{13}$$

$$T_e = \frac{p}{w_{syn}}\left(\psi_{qr}i_{dr} + \psi_{dr}i_{qr}\right) = \frac{p}{w_{syn}}\left(\psi_{ds}i_{qs} + \psi_{qs}i_{ds}\right) \tag{14}$$

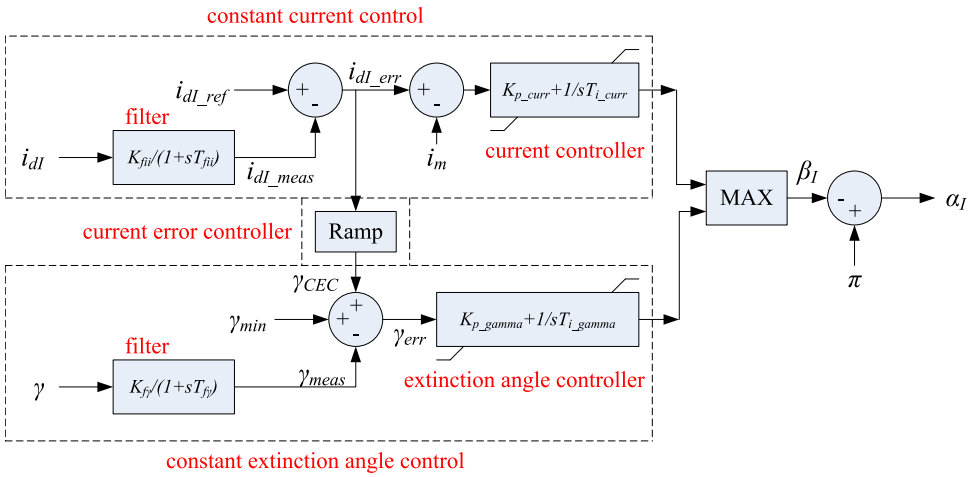


Fig. 4. Inverter control model.

$$u_{rd} = \frac{\sqrt{3}}{2\sqrt{2}} PWM_d \frac{U_{DC}}{U_{rnom}} \tag{15}$$

$$u_{rq} = \frac{\sqrt{3}}{2\sqrt{2}} PWM_q \frac{U_{DC}}{U_{rnom}} \tag{16}$$

$\underline{u}_s$,$\underline{i}_s$,$\underline{\psi}_s$: space vectors for stator voltage, current and flux

$\underline{u}_r$,$\underline{i}_r$,$\underline{\psi}_r$: space vectors for rotor voltage, current and flux

w_{syn},w_r,w_n: synchronous angular velocity, angular velocity of rotor and niminal electrical frequency of network

R_s,R_r: stator resistance and rotor resistance

p: pole pair

T_e: electromagnetic torque

ψ_{qs},ψ_{ds},ψ_{qr},ψ_{dr}: q-axis stator flux, d-axis stator flux, q-axis rotor flux and d-axis rotor flux

i_{qs},i_{ds},i_{qr},i_{dr}: q-axis stator current, d-axis stator current, q-axis rotor current and d-axis rotor current

u_{rd},u_{rq}: d-axis rotor voltage and q-axis rotor voltage

PWM_d,PWM_q: d-axis modulation index and q-axis modulation index

U_{DC},U_{rnom}: DC link voltage and nominal rotor voltage

2.2.2.2. Wind turbine mechanism. In Fig. 5, it shows the model adopted for wind turbine mechanism, which is a lumped mass-damping-spring model with two masses. The large mass and the small mass correspond to the wind turbine inertia and the generator rotor inertia, respectively. For the shaft, only the low-speed shaft is considered, and the high-speed shaft is assumed to be rigid. The dynamic behaviors are described by the following equations.

$$\frac{d\theta_{wt}}{dt} = w_{wt} \tag{17}$$

$$\frac{d\theta_{gen}}{dt} = w_{gen} \tag{18}$$

$$J_{wt}\frac{dw_{wt}}{dt} = T_{wt} - T_{shaft} \tag{19}$$

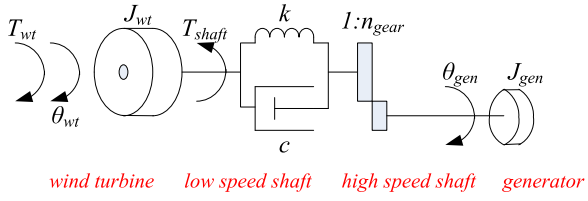


Fig. 5. Model of wind turbine mechanism.

$$J_{gen}\frac{dw_{gen}}{dt} = -T_e + n_{gear}T_{shaft} \tag{20}$$

$$T_{shaft} = c\left(w_{wt} - \frac{w_{gen}}{n_{gear}}\right) + k\left(\theta_{wt} - \theta_{gen}/n_{gear}\right) \tag{21}$$

k,c: stiffness and damping coefficience of shaft

n_{gear}: gear ratio

T_{wt},T_{shaft}: torques of wind turbine and shaft

T_e: electromagnetic torque

θ_{wt},θ_{gen}: angles of wind turbine and generator

w_{wt},w_{gen}: angular velocities of wind turbine and generator

2.2.2.3. DFIG control. The control scheme of DFIG is shown in Fig. 6. There is an outer loop to regulate the active and reactive power, and an inner loop to regulate the active and reactive current. By regulating the q- and d-axis rotor currents, respectively, the active and reactive power can be controlled. The dynamic equations can be obtained as the followings.

$$i_{q_ref} = K_P\left(P_{grid_ref} - P_{grid_meas}\right) + \frac{1}{T_P}\int\left(P_{grid_ref} - P_{grid_meas}\right)dt \tag{22}$$

$$PWM_q = K_q\left(i_{q_ref} - i_q\right) + \frac{1}{T_q}\int\left(i_{q_ref} - i_q\right)dt \tag{23}$$

$$i_{d_ref} = K_Q\left(Q_{grid_ref} - Q_{grid_meas}\right) + \frac{1}{T_Q}\int\left(Q_{grid_ref} - Q_{grid_meas}\right)dt \tag{24}$$

$$PWM_d = K_d\left(i_{d_ref} - i_d\right) + \frac{1}{T_d}\int\left(i_{d_ref} - i_d\right)dt \tag{25}$$

P_{grid_meas},Q_{grid_meas}: measured active power and measured reactive power

P_{grid_ref},Q_{grid_ref}: reference active power and reference reactive power

i_d,i_q: d-axis rotor current and q-axis rotor current

2.2.3. Modeling of steam turbine generator unit

2.2.3.1. Synchronous generator. The synchronous generator is modeled with the detailed 6-order model, and the IEEE Type1 AVR (Automatic Voltage Regulator) is included. Those models are common and will be not repeated here.

2.2.3.2. Steam turbine mechanism. The turbine-and-generator mechanism needs to be included in order for the studies on the turbine shafts dynamics, hence the multi-mass model is adopted. The turbine-and-

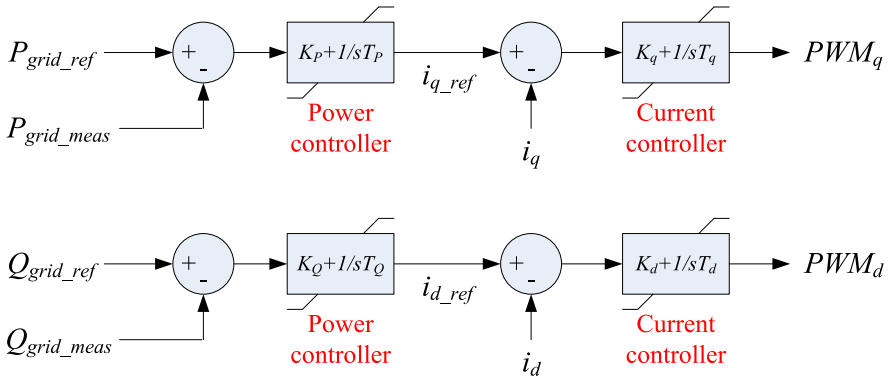


Fig. 6. DFIG control model.

generator mechanism of each unit studied is equipped with a high pressure turbine (HP), an intermediate pressure turbine (IP), two low pressure turbines (LPA and LPB) and an exciter (EXC), in additional to the generator rotor (GEN). The lumped mass-damping-spring model of the 6-mass model is shown in Fig. 7, of which the model parameters are listed in Table 1. The dynamic behaviors can be described by the following equations.

$$T_i - K_{i-1_i}(\varphi_i - \varphi_{i-1}) - K_{i_i+1}(\varphi_i - \varphi_{i+1}) - D_i w_i = J_i \frac{dw_i}{dt} \tag{26}$$

$$w_i = \frac{d\varphi_i}{dt} \tag{27}$$

T: turbine torque
J: turbine inertia
K: shaft stiffness
D: turbine damping
w: turbine angular velocity
φ: turbine angle
i: turbine mass (i.e. HP, IP, LPA, LPB, GEN and EXC)
$i-1$: previous turbine mass
$i+1$: next turbine mass
T_e: electromagnetic torque

2.2.4. Modal analysis method

By linearizing the dynamic equations of the entire power system studied around the operation point, the behavior of the system can be represented with the differential equations:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} \tag{28}$$

$\mathbf{x}$: column vector of state variables
$\mathbf{A}$: characteristic matrix

Left and right eigenvectors, which reflect the observability and the controllability of the system, that correspond to eigenvalue λ_i are defined by expressions:

$$\mathbf{A}\mathbf{r}_i = \lambda_i \mathbf{r}_i \tag{29}$$

$$\mathbf{l}_i\mathbf{A} = \mathbf{l}_i\lambda_i \tag{30}$$

Table 1
Parameters of the mass-damping-spring model.

Parameter	G1 unit	G2 unit
J_{HP}[kgm^2]	1,166.619	1,166.619
K_{HP_IP}[Nm/rad]	38,077,813	45,693,376
J_{IP}[kgm^2]	1,953.917	2,051
K_{IP_LPA}[Nm/rad]	68,902,240	82,682,688
J_{LPA}[kgm^2]	10,783.35	10,783.35
K_{LPA_LPB}[Nm/rad]	102,652,087	123,182,504
J_{LPB}[kgm^2]	11,104.15	11,104.15
K_{LPB_GEN}[Nm/rad]	139,777,120	167,732,544
J_{EXC}[kgm^2]	429.6976	429.6976
K_{EXC_GEN}[Nm/rad]	5,562,837	6,675,404

λ_i: eigenvalue of matrix
$\mathbf{A}\mathbf{r}_i$,$\mathbf{l}_i$: right column eigenvector and left row eigenvector.

The participation factor can reflect the relative contribution of a given state variable to the modes. For the participation factor of k-state variable, it is determined by the formula:

$$p_i^k = |l_i^k r_i^k| \tag{31}$$

l_i^k,r_i^k: the k-th elements of the left and right eigenvectors associated with the i th mode

3. Preliminary studies

Before incorporation of the wind farm into the system, the preliminary studies are made to understand the system behaviors by both the modal analysis and transient simulations. Under the preliminary studies, the generator units G1 and G2 operate with rated power of 800 MW, and the HVDC link is controlled to transmit 1000 MW of power. The excess power is dissipated in local load.

3.1. Modal analysis

By applying the modal analysis to the system studied, the controllability, observability and participation factor can be calculated. For the system without wind farm, there are 53 state variables and 17

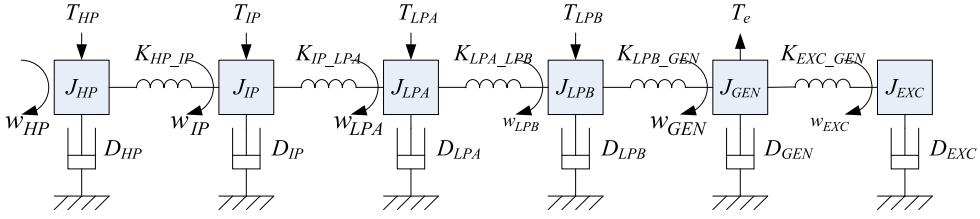


Fig. 7. Lumped mass-damping-spring model of turbine-and-generator mechanism.

eigenvalues with complex conjugate in total. However, it is focused only on the torsional vibration modes of turbine generators and the states variables associated with HVDC converter controls. In Table 2, it shows the torsional modes of the two generator units. Each generator unit has 5 torsional modes. It can be seen that each torsional mode has positive damping ratio, so every mode is dynamics-stable.

In Fig. 8, it shows the participation factors for various torsional modes. However, only the state variables corresponding to converter controls are presented, because it is aimed for SSTI studies. The states 24 to 28 are associated with the inverter controls, and the states 29 to 33 are associated with the rectifier controls. It can be seen that the mode 1 has the biggest participation factors, and almost all the other modes have negligible ones. So, it will be evident that the mode 1 will be the most affected one when subjecting to SSTI with HVDC controls. Moreover, the interaction with the rectifier controls will be more significant than with the inverter controls.

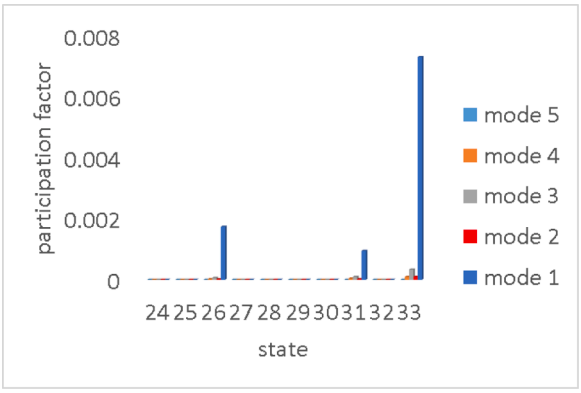


Fig. 8. Participation factors.

3.2. Transient simulations

The time-domain torsional vibrations can be simulated by using the electromagnetic transient simulation function provided by the DIgSILENT PowerFactory. It is assumed that a three-phase short-circuit fault is applied to the 345 kV bus at the rectifier side of HVDC transmission system, and is cleared after 0.1 s (or 5 cycles). The fault impedance is 0 + j75 Ω. The resulting responses are shown in Fig. 9.

In Fig. 9(a), (b), (c), and (d), respectively, they show the responses of firing angles, rectifier side AC bus, inverter side AC bus and DC line terminals. It can be seen that the firing angle of rectifier reduces to the minimum value of 5° during faulting period, but the rectifier controller is unable to regulate the current to its set-point. The voltages and currents of the rectifier AC bus, the inverter AC bus and the DC line terminal drop to about 0.8 pu. The inverter controller switches to current-control mode. When the fault clears, the rectifier controller takes over current control again. After some time, the inverter controller switches back to extinction angle control. In Fig. 9(e), it shows the responses of torsional angles for various shafts of G1 units. It can be seen that the responses are stable, and the main vibration composition is the mode 1.

4. Incorporation of the wind farm

In this section, the wind farm is incorporated into the system. Since the rated power of the HVDC transmission system is limited to 1000 MW, the augmentation in power due to the incorporation of wind farm will be counterbalanced by the generator unit G1. So, the G1 needs to operate with de-rated power. It is found in the followings that the turbine shafts torsional responses of the G1 unit are significantly affect by the generator de-rating operations.

4.1. Modal analysis

For the modal analysis, the seven DFIG wind farm models are incorporated into the system step by step. The generation power of the G1 decreases from 800 MW to 100 MW by an interval of 100 MW following the power of wind farm increases from 0 MW to 700 MW by an interval of 100 MW.

Table 2
Torsional modes.

	G1 unit		G2 unit	
Mode	frequency (Hz)	damping ratio	frequency (Hz)	damping ratio
1	13.3	0.00404	14.5	0.003942
2	18.3	0.000262	20.1	0.000242
3	22.7	0.005837	24.8	0.00527
4	28.4	0.001304	31.1	0.0012
5	43.3	0.002054	46.7	0.001951

In Fig. 10, it shows the damping ratios of various torsional modes for different generation power of G1. It can be seen that all of the torsional modes are negatively affected. The damping ratios decrease following the decrease in generation power. Among them, the mode 3 is most affected, though it is not the main vibration mode. The mode 1 comes next though it is the main vibration mode for various shafts, and following that is the modes 4 and 5. The mode 2 is nearly unaffected though it has the smallest damping ratio in rated power operation.

For the mode 3, the damping ratio decreases from 0.006 to −0.006 following the generation power decreases from 800 MW to 100 MW. At about 450 MW of generation power, the damping ratio starts to change from positive to negative. Since the mode 3 is relatively significant for every shaft, except the LPA_LPB shaft, the stability of turbine shafts torsional responses will be significantly affected by the generator de-rating operations.

For the modes 1, 2 and 4, no matter what the degree of affect is, the damping ratios keep positive. So, even though the mode 1 is the main vibration mode for every shaft in rated power operation, it will be no longer important when concerning about the de-rating operations.

For the mode 5, the damping ratio decreases from 0.002 to −0.001 following the generation power decreases from 800 MW to 100 MW. At about 400 MW of generation power, the damping ratio starts to change from positive to negative. The mode 5 is possible to become unstable following the generator de-rating operations, however, it will be not so significant because it plays just a negligible role in shafts torsional vibrations.

To sum up the modal analysis, the effect of generator de-rating operations is to introduce negative damping into the torsional modes. The effect on torsional vibrations and stability is negative, and the mode 3 is the most affected one.

4.2. Transient simulations

The modal analysis has shown that the dynamic stability of turbine shafts responses might be affected by the generator de-rating operations. For the responses subjecting to a large disturbance, transient simulations are also made to find their effect. It is assumed that the same events are applied. The cases of integrating 300 MW and 500 MW of wind power are studied.

In Fig. 11, it shows the torsional vibrations of the G1 unit for incorporation of 300 MW and 500 MW of wind farm power. The responses without wind farm (i.e. G1 operates with rated power of 800 MW) are also plotted together in order for comparing. For the case of integrating a wind farm of 300 MW (i.e. G1 operates with de-rated power of 500 MW), it can be seen that the vibrations for the de-rated power operation are severer than the rated power operation. This is somewhat weird, but can be easily understand by further checking the frequency spectrum of responses. It can be found that the mode 1 is dominant under the rated power operation condition, however, it turns

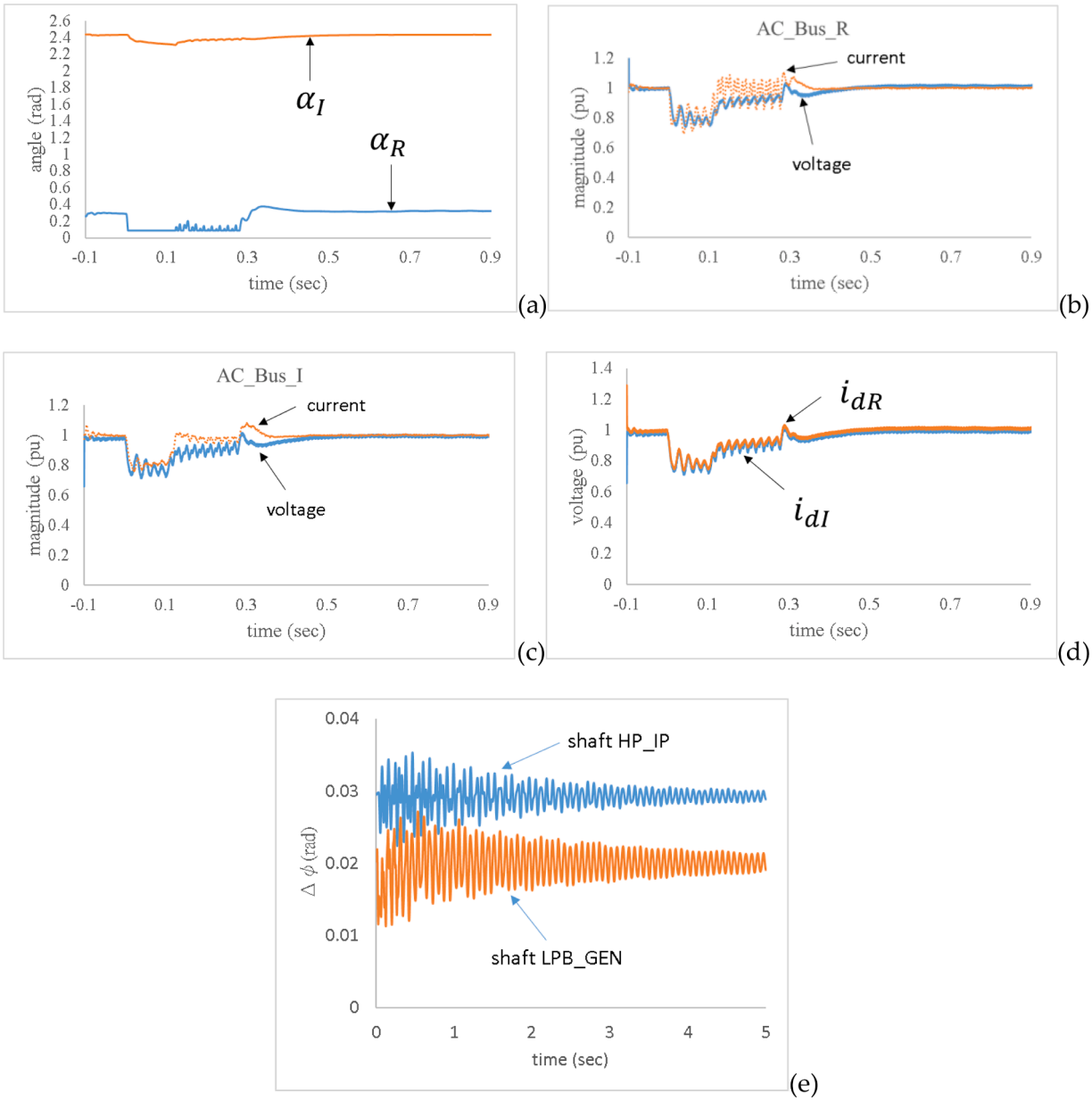


Fig. 9. Transient responses, (a) the firing angles, (b) the rectifier side AC bus, (c) the inverter side AC bus, (d) the DC line terminal and (e) the torsional vibrations of G1 unit.

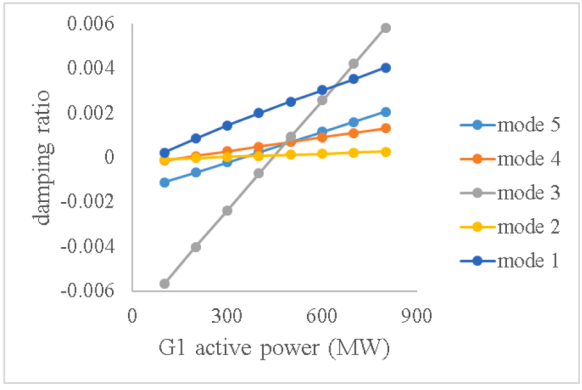


Fig. 10. Damping ratios of torsional modes under various G1 generation power situations.

to be the mode 3 to dominate the vibration behavior under the de-rated power operation condition. Since the damping ratio of mode 3 decreases rapidly under the de-rating operations as revealed in Section 4.1., the vibrations will last much more longer so that the responses show severer vibrations.

For the case of integrating a wind farm of 500 MW (i.e. G1 operates with de-rated power of 300 MW), it can be seen that the responses at the shaft HP_IP and the shaft LPB_GEN are divergent. So, it shows that the generator de-rating operation could lead to transient instability in torsional vibrations. And the unstable composition is the mode 3, just as the modal analysis reveals.

To sum up the transient simulations, a small degree of de-rating operation could make shafts torsional vibrations become more serious. A large degree of de-rating operation could de-stabilize shafts torsional responses.

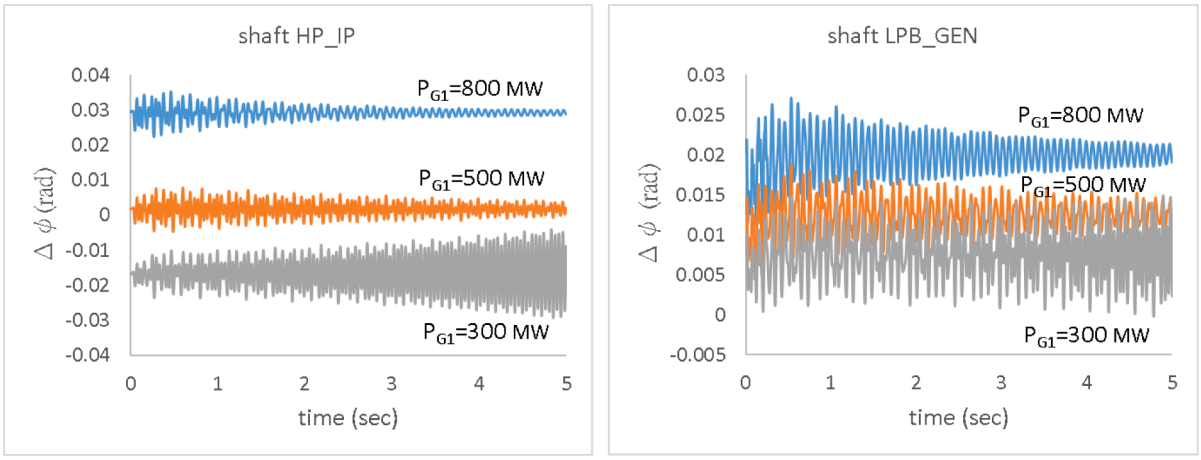


Fig. 11. Torsional vibrations for incorporation of a wind farm with different power, (a) shaft HP_IP and (b) shaft LPB_GEN.

5. SSTI with HVDC controls

Since the de-rating operations of turbine generators have been proved to be capable of significantly changing torsional response characteristics it is further studied in this section whether the SSTI with HVDC controls will be affected or not. Firstly, participation factor is studied to make a preliminary judgement. In Fig. 12, it shows the comparison of participation factors before and after incorporation of a wind farm of 700 MW (i.e. G1 operates with 800 MW and 100 MW, respectively). Since the mode 1 is the most sensitive to SSTI, it is studied first. In addition, the mode 2 is also studied though it has the far smaller participation factor than the mode 1, because it is found the effect cannot be ignored. According to the charts, the participation factors corresponding to de-rating operations are increased for both the mode 1 and the mode 2. That implies the SSTI with HVDC controls will become more significant after the incorporation of wind farm (or generator operating with de-rated power). The modes 3, 4 and 5 are dismissed due to few or none interactions with HVDC controls.

5.1. Effect of the rectifier controls

The main control function of rectifier is achieved by the PI controller which regulates the DC current. The nominal settings of this PI controller are $K_p = 1$ for the proportional gain and $T_i = 0.01$ for the integral time constant. For studying the effect of control parameters on torsional interactions, eigen-analysis is adopted and the damping ratios of torsional modes are focused. The K_p value is assumed to be varied between 1 and 3, and the T_i value is assumed be varied between 0.01 and 1. In Fig. 13, it shows the effect of K_p on torsional modes of G1 unit. It can be seen that the mode 1 is most affected, the mode 2 comes next, and other modes are unaffected. This result is consistent with the result of participation factor analysis. So, only the modes 1 and 2 are focused in the following studies.

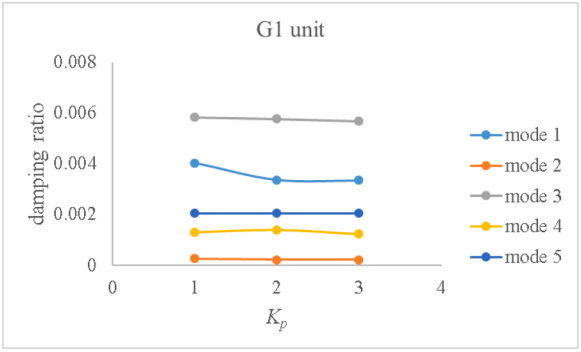


Fig. 13. Effect of proportional gain of the rectifier controller on torsional modes.

5.1.1. Effect of K_p

In Fig. 14(a), it shows the effect of K_p settings on damping ratios of modes 1 and 2 under various generator de-rating operation situations. It can be seen that there is indeed significant effect.

For the mode 1, under the 800 MW situation, the damping ratio decreases from 0.00404 to 0.003348 following the increases of K_p from 1 to 3. The difference is 0.000692. Under the 100 MW situation, the damping ratio decreases from 0.00021 to −0.00108 following the increases of K_p from 1 to 3. The difference is 0.00129. By comparing the

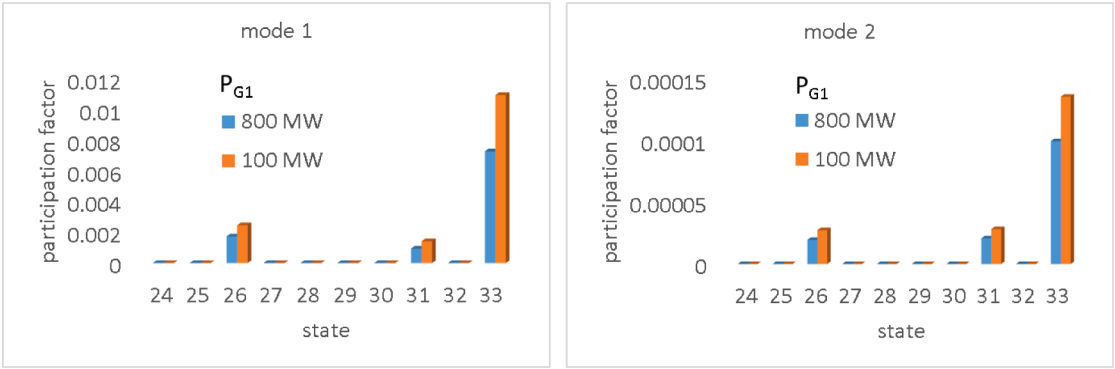
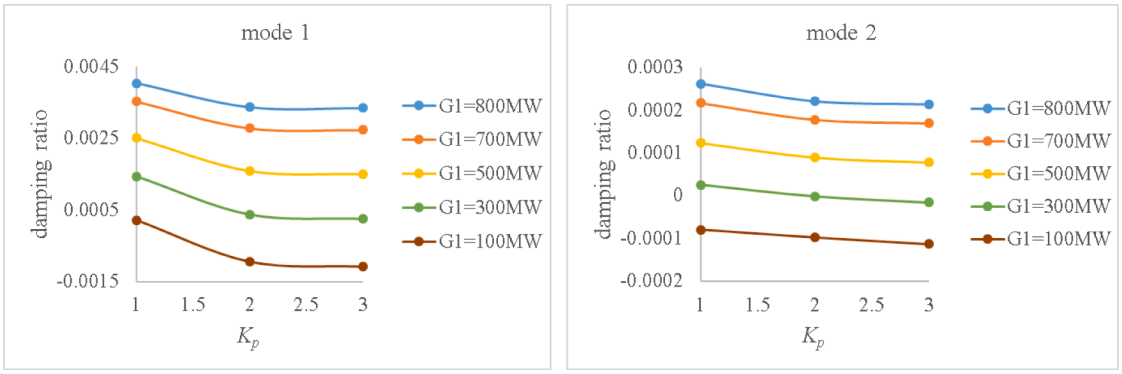
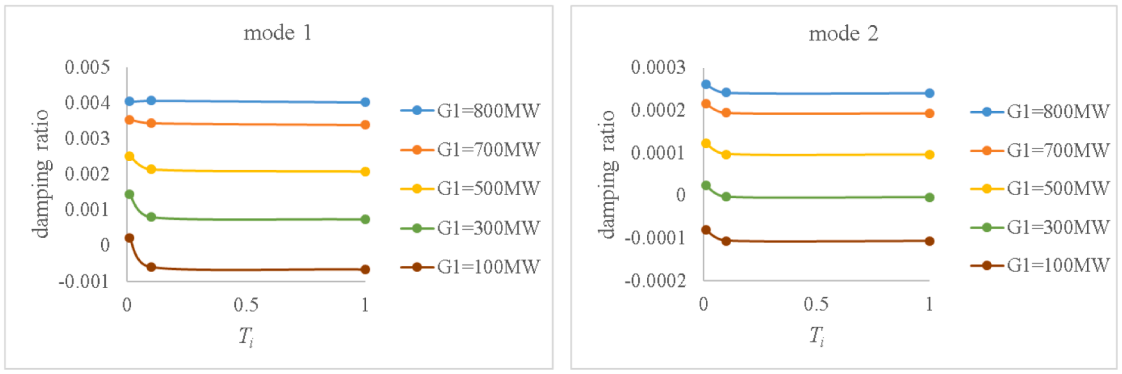


Fig. 12. Comparison of participation factors.



(a) effect of proportional gain (K_p).



(b) effect of integral time constant (T_i).

Fig. 14. Effect of rectifier control parameters on torsional modes damping ratios.

two situations, it is obvious that the influence of K_p setting becomes bigger under the de-rated power operation situations.

For the mode 2, under all the generation power situations, the damping ratios decrease by the same slope following the increases of K_p from 1 to 2. The damping ratio is no longer decreased for K_p increasing from 2 to 3 under the 800 MW situation. In contrast, the damping ratio continues to decrease for K_p increasing from 2 to 3 under the 100 MW situation. So, the influence of K_p setting also becomes bigger under the de-rated power operations.

Furthermore, it is noteworthy that the negative effects of SSTI with K_p and generator de-rating operations are cumulative. The double action of the two effects could make the situation worse. For the mode 1, it should be paid attention for the situations of generation power lower than 300 MW because the mode could be de-stablized. For example, under the 200 MW situation, the mode would be stable when setting K_p to 1, and unstable when setting K_p to 3. For the mode 2, attention should be paid on the situation that the generator generation power is less than 300 MW. For example, under the 300 MW situation, the mode would get stable when setting K_p to 1, and unstable when setting K_p to 3.

Moreover, it is noteworthy to make a comparison of influence level between the effects of SSTI with K_p and generator de-rating operations. Let's focus on the mode 1 as an example. Under the normal operation of 800 MW, the difference in damping ratio is just 0.000692 following the increases of K_p from 1 to 3. The 100 MW operation is the worst situation, of which the difference increases to 0.00129. In contrast, for $K_p = 1$, the difference in damping ratio is 0.00383 for operation power decreasing from 800 to 100 MW. For $K_p = 3$, the difference is even increased up to 0.004428. So, the effect of SSTI with K_p is far smaller than the effect of generator de-rating operations. Roughly speaking, the effect of SSTI with K_p is just equivalent to the effect of 100 MW reduction in operation power.

5.1.2. *Effect of* T_i

In Fig. 14 (b), it shows the effect of T_i settings on damping ratios of modes 1 and 2 under various generator de-rating operation situations. It can also be seen that there is significant effect.

For the mode 1, under the 800 MW situation, the damping ratio almost doesn't change following the increases of T_i from 0.01 to 1. That is there is nearly no SSTI with T_i. Under the 100 MW situation, the damping ratio decreases from 0.000214 to −0.00065 following the increases of T_i from 0.01 to 1. The difference is 0.000864. By comparing the two situations, it is obvious that the de-rating operation makes the mode 1 changing from non-SSTI with T_i to negative SSTI effect. So, the influence of T_i setting becomes bigger under the de-rated power operation situations.

For the mode 2, under all the generation power situations, the damping ratios decrease by the same slope following the increases of T_i from 0.01 to 1. That means the generator de-rating operations won't affect the effect of SSTI with T_i.

Furthermore, the negative effects of SSTI with T_i and generator de-rating operations are also cumulative. It is necessary to pay attention to the double action of the two effects. For the mode 1, it should be careful when the generation power is less than 200 MW. For example, under the 100 MW situation, the mode would be stable when setting T_i to 0.01, and unstable when setting T_i to 0.1. For the mode 2, attention should be paid on the situation that the generator generation power is less than 300 MW. For example, under the 300 MW situation, the mode would be stable when setting T_i to 0.01, and unstable when setting T_i to 0.1.

The comparison of influence level between the effects of SSTI with T_i and generator de-rating operations can be further made with respect to

the mode 1. For the worst generation power (i.e. 100 MW) condition, the damping ratio decreases 0.000864 following the increases in T_i. In contrast, for $T_i = 0.01$, the difference in damping ratio is 0.003826 for operation power decreasing from 800 to 100 MW. For $T_i = 1$, the difference is increased up to 0.004678. So, the effect of SSTI with T_i is far smaller than the effect of generator de-rating operations. Roughly speaking, the effect of SSTI with T_i is about equivalent to the effect of 150 MW reduction in operation power.

To sum up the SSTI with rectifier controls, the most sensitive mode to the SSTI effect is the mode 1. The generator de-rating operation could make the effect of SSTI with rectifier controls become more negative, and even change from non-SSTI effect to negative SSTI effect. Furthermore, the effect of SSTI with rectifier controls and the effect of generator de-rating operations are cumulative, and the effect of SSTI with rectifier controls is far smaller than the effect of generator de-rating operations.

5.2. Effect of the inverter controls

The main control function of inverter is achieved by the PI controller which regulates the extinction angle. The nominal settings of this gamma controller are $K_{p_gamma} = 0.1$ for the proportional gain and $T_{i_gamma} = 0.01$ for the integral time constant. For studying the effect of control parameters on torsional interactions, the K_{p_gamma} value is varied between 0.1 and 0.4, and the T_{i_gamma} value is varied between 0.01 and 1. In Fig. 15, it shows the effect of K_{p_gamma} on torsional modes of G1 unit. It can be seen that the mode 1 is most affected, and other modes are almost unaffected. This result is also consistent with the result of participation factor analysis. So, only the modes 1 and 2 are discussed in the following studies.

5.2.1. Effect of K_{p_gamma}

In Fig. 16(a), it shows the effect of K_{p_gamma} settings on damping ratios of modes 1 and 2 under various generator de-rating operation situations. It can be seen that the damping ratios of modes 1 and 2 are affected by the K_{p_gamma} settings, and the effect of SSTI with K_{p_gamma} is influenced by the generator de-rating operations as well.

For the mode 1, under the 800 MW situation, the damping ratio shows a droop curve following the increases in K_{p_gamma} value. Under the 100 MW situation, the damping ratio curve first rises then falls. Obviously, the effect of SSTI with K_{p_gamma} has been changed by the de-rating operations, and this change is somewhat irregular. The difference in damping ratio is 0.00243 under the 800 MW situation, and is 0.002768 under the 100 MW situation, for K_{p_gamma} changing from 0.1 to 0.4. So, the influence of K_{p_gamma} setting becomes a little bigger under the de-rated power operation situations.

For the mode 2, the damping ratio curves affected by K_{p_gamma} are irregular as well. The generation power of 600 MW is the dividing line. When the generation power is greater than 600 MW, the increase in K_{p_gamma} value would lead to the increase in damping ratio. The effect is positive. When it is less than 600 MW, the increase in K_{p_gamma} would lead to the decrease in damping ratio. The effect becomes negative. So, the de-rating operation makes the SSTI of mode 2 with K_{p_gamma} changing from positive effect to negative effect.

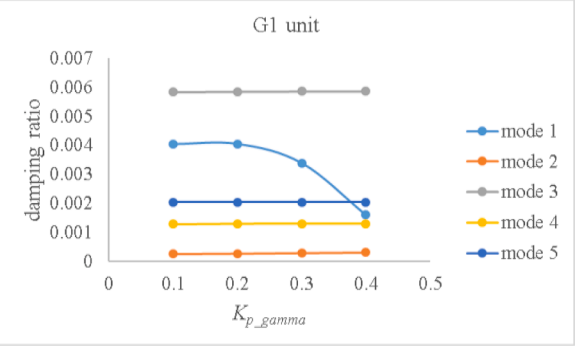


Fig. 15. Effect of proportional gain of the inverter controller on torsional modes.

Furthermore, the effect of SSTI with K_{p_gamma} (no matter positive or negative) and the effect of generator de-rating operations are found to be cumulative, too. So, attention should be paid on the situation that the generation power is less than 500 MW for the mode 1, and on the situation that the generation power is less than 300 MW for the mode 2. Under those situations, the modes are possible to become unstable if K_{p_gamma} is improperly set.

The comparison of influence level between the effects of SSTI with K_{p_gamma} and generator de-rating operations is further made with respect to the mode 1. For K_{p_gamma} changing from 0.1 to 0.4, the differences of damping ratio are 0.00243 and 0.002768, respectively, for the 800 and 100 MW power situations. In contrast, for $K_{p_gamma} = 0.1$, the difference in damping ratio is 0.003844 for operation power decreasing from 800 to 100 MW. For $K_{p_gamma} = 0.4$, the difference is increased to 0.004182. So, the effect of SSTI with K_{p_gamma} is still smaller than the effect of generator de-rating operations, however, is very close. Roughly speaking, the effect of SSTI with K_{p_gamma} can be equivalent to the effect of 500 MW reduction in operation power.

5.2.2. Effect of T_{i_gamma}

In Fig. 16(b), it shows the effect of T_{i_gamma} settings on damping ratios of modes 1 and 2 under various generator de-rating operation situations. It can be seen that the effect is also significant.

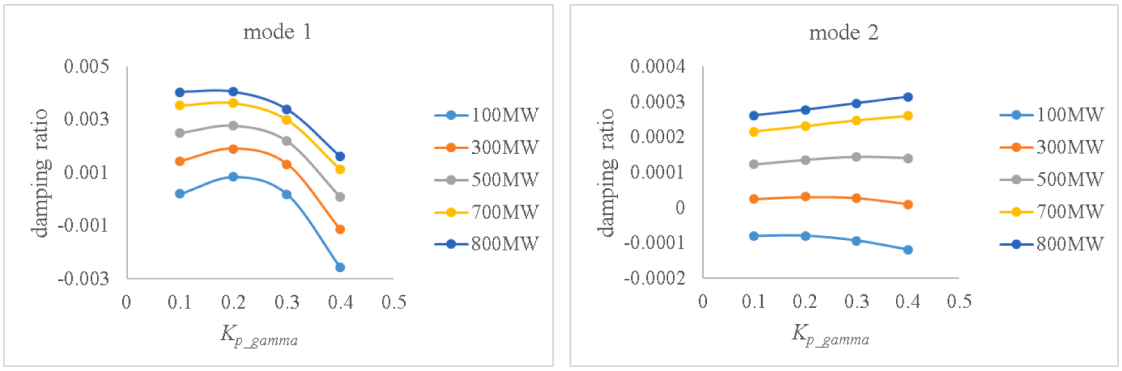
For the mode 1, under the 800 MW situation, the damping ratio decreases from 0.00404 to 0.003762 following the increases of T_{i_gamma} from 0.01 to 1. The difference is 0.000278. Under the 100 MW situation, the damping ratio decreases from 0.0000198 to −0.00029 following the increases of T_{i_gamma} from 0.01 to 1. The difference is 0.000488. By comparing the two situations, it is obvious that the influence of T_{i_gamma} setting becomes bigger under the de-rated power operation situations.

For the mode 2, under the 800 MW situation, the damping ratio shows a droop curve following the increases in T_{i_gamma} value. The effect is negative. Under the 100 MW situation, the damping ratio curve shows a rising curve. The effect is positive. Obviously, the effect of SSTI with T_{i_gamma} has been changed from negative to positive by the de-rating operations. However, this change can be ignored due to very small in magnitude.

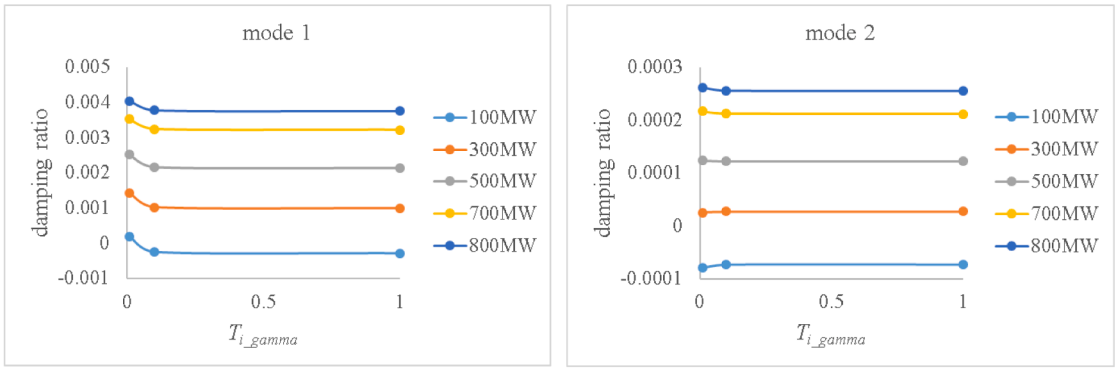
Furthermore, the effect of SSTI with T_{i_gamma} (no matter positive or negative) and the effect of generator de-rating operations are cumulative. However, only the situation of 100 MW generation power needs to be paid attention, under which the mode 1 might become unstable if T_{i_gamma} is improperly set.

For the mode 1, the comparison of influence level between the effects of SSTI with T_{i_gamma} and generator de-rating operations is further made. Under the worst situation (i.e. 100 MW generation power), the damping ratio decreases 0.000488 for T_{i_gamma} increasing from 0.01 to 1. In contrast, for $T_{i_gamma} = 0.01$, the difference in damping ratio is 0.003842 for operation power decreasing from 800 to 100 MW. For $T_{i_gamma} = 1$, the difference is increased to 0.004052. So, the effect of SSTI with T_{i_gamma} is far smaller than the effect of generator de-rating operations. Roughly speaking, the effect of SSTI with T_{i_gamma} is less than the effect of 100 MW reduction in operation power.

To sum up the SSTI with inverter controls, the most sensitive mode to the SSTI effect is the mode 1. The generator de-rating operation could make the effect of SSTI with inverter controls become more significant, and even change from positive effect to negative effect or vice versa. Furthermore, the effect of SSTI with inverter controls and the effect of generator de-rating operations are cumulative, and the effect of SSTI with inverter controls is smaller than (but can be close to) the effect of generator de-rating operations. However, it must be noted that the comparison of influence level is made with respect to the mode 1 alone.



(a) effect of proportional gain (K_{p_gamma}).



(b) effect integral time constant (T_{i_gamma}).

Fig. 16. Effect of inverter gamma control parameters on torsional modes damping ratios.

If the mode 3 is considered, the effect of generator de-rating operations will be totally dominant.

6. Discussions on interactions between adjacent units

Since there is a generator unit G2 neighboring to the G1, it is necessary to know how will the de-rating operation of G1 unit affect the G2 unit. For this purpose, modal analysis is made to study the effect of G1 de-rating operations on the SSTI between G2 torsional modes and HVDC controls. However, only the mode 1 is focused because it is the most sensitive one. In Fig. 17, it shows the damping ratios of the G2 torsional mode 1 subjecting to changes in K_{p_gamma} under various G1 generation power situations. It can be seen that the effect is significant.

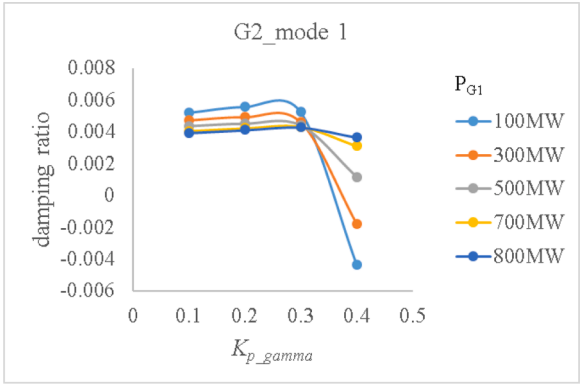


Fig. 17. Damping ratios of the G2 torsional mode 5 subjecting to changes in K_{p_gamma} under various G1 generation power situations.

When the K_{p_gamma} value is between 0.1 and 0.3, the damping ratios get larger for G1 operates with de-rated power. The effect is positive. However, when the K_{p_gamma} value is greater than 0.3, the damping ratios turn to drop. Under the 800 MW situation, the damping ratio drops from 0.003943 to 0.003665. the difference is 0.000278. Under the 100 MW situation, the damping ratio drops from 0.005221 to −0.00433. the difference is 0.009551. By comparing the two situations, it is evident that the G1 de-rating operations could largely exacerbate the SSTI effect between the HVDC controls and the G2 torsional modes. And the effect is negative.

7. SSTI with DFIG controls

The DFIG control function is mainly composed of the current controls and the power controls. The main control function of current controls is achieved by a PI controller of which the nominal settings are $K_d = K_q = 0.0496$ for the proportional gain and $T_d = T_q = 0.0128$ for the integral time constant. The main control function of power controls is achieved by a PI controller of which the nominal settings are $K_P = K_Q = 4$ for the proportional gain and $T_P = T_Q = 0.1$ for the integral time constant. For studying the effect of control parameters on torsional interactions, eigen-analysis is made and damping ratios of torsional modes are focused. The proportional gain of current controller is assumed to be varied between 0.01 and 0.45, and the integral time constant is assumed be varied between 0.003 and 0.055. The proportional gain of power controller is assumed to be varied between 1 and 12, and the integral time constant is assumed be varied between 0.01 and 1. And it is assumed that the DFIG wind farm of 500 MW is incorporated into the system.

In Fig. 18, it shows the resulting damping ratios of torsional modes of

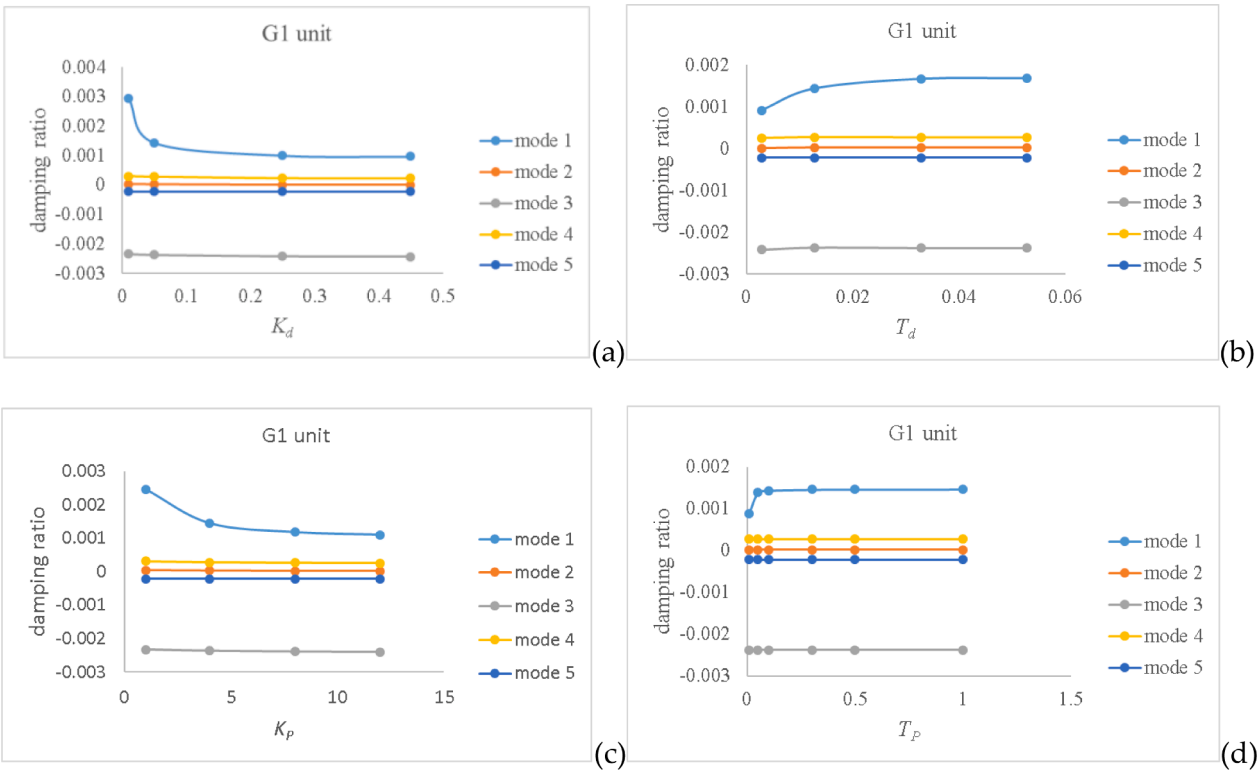


Fig. 18. Torsional modes damping ratios influenced by (a) proportional gain of current controller, (b) integral time constant of current controller, (c) proportional gain of power controller and integral time constant of power controller.

G1 for varying in parameter values. It can be seen that the mode 1 is sensitive to SSTI with DFIG controls, while other modes have little or no effect. This result is consistent with the result of participation factor analysis, though which is not shown here. So, only the mode 1 is focused in the following studies.

For studying the influence of generator de-rating operations on the effect of SSTI with DFIG controls, analyses have been made for some cases in which different wind power is introduced. For the influence on SSTI with proportional gain of current controller, it is shown in Fig. 19, of which (a) is for the G1 unit and (b) is for the G2 unit. For the G2 unit, it has been operating at rated power and does not change due to the incorporation of wind farm. So, the changes in the SSTI effect shown in the figure is caused purely by the wind farm. In contrast, the G1 unit will operate with de-rated power to counterbalance the power of wind farm. So, the changes in the SSTI effect shown in the figure is caused not only by the wind farm but also by the de-rating operations. From these two figures, it can be observed that the more the power of wind farm, the greater the effect of SSTI with proportional gain of current controller. The effect of generator de-rating operation will accumulate on the SSTI effect, but will not change the effect of such a SSTI. In order to make it easier to distinguish this effect, the influence curve of SSTI on mode 1 under different operation power for the G2 unit is shown in Fig. 20. For this studies, the G2 unit is dispatched to operate with reduced power, and the power of wind farm is kept at 500 MW without change. It is easy to see from the graph that the relationship curves are parallel, indicating that the trend of influence will not change but only accumulate.

For the influences on SSTI with integral time constant of current controller, with proportional gain of power controller and with integral time constant of power controller, they are shown in Figs. 21, 22 and 23, respectively. From these figures, it can be found that the result is the same as above. Overall, generator de-rating operation would not change the effect of SSTI with DFIG controls, but it does have a cumulative effect. As compared with the influence of generator de-rating operation on SSTI with HVDC controls, the influence of generator de-rating operation on SSTI with DFIG controls is more simple.

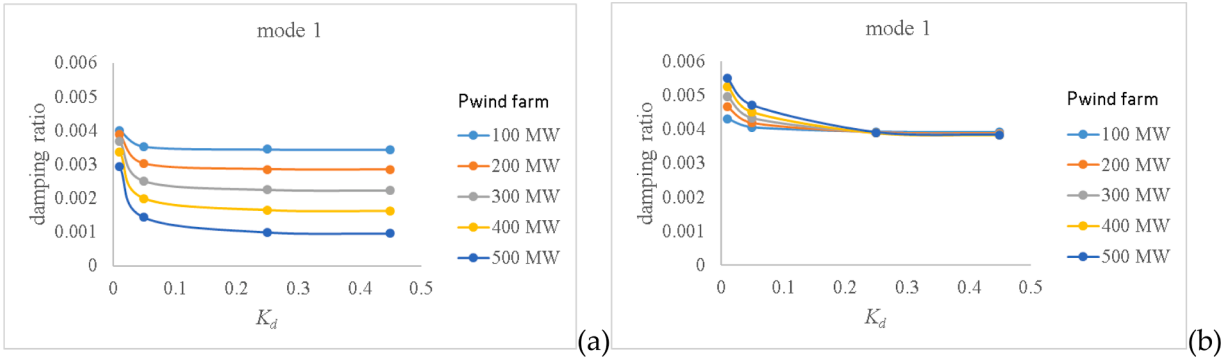


Fig. 19. The influence of generator de-rating operations on the effect of SSTI with proportional gain of current controller for (a) the G1 unit and (b) the G2 unit.

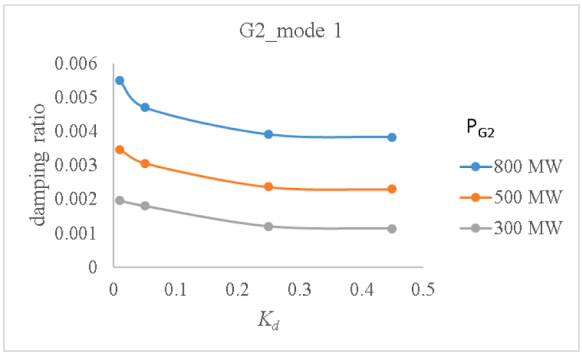


Fig. 20. The influence curve of SSTI on mode 1 of G2 unit under different power operation of G2 unit.

8. Discussions on SCR (Short circuit ratio)

In general, the HVDC system should prevent from connecting to a very weak AC network, otherwise the HVDC system might not operate smoothly. In other words, the SCR should not be lower than 1.5. So, it is necessary to study the effect of SCR on the SSTI with HVDC controls and on the SSTI with DFIG controls. For this purpose, the rectifier side of SCR is varied from 1.8 to 2.7 and 3.9 by increasing the number of generator units and loads. Eigen-analysis is made for each SCR to study the SSTI under the situation of integration of a wind farm of 500 MW.

For the results of the SSTI with HVDC controls, only the effect of SCR on SSTI with proportional gain of rectifier controller is shown in Fig. 24 (a), the effect on others is not shown because the phenomenon is almost the same. It can be seen from the figure that the increase in SCR reduces the role of SSTI. For the results of the SSTI with DFIG controls, only the effect of SCR on SSTI with proportional gain of current controller is shown in Fig. 24(b), the effect on others is not shown, either. Also, it be

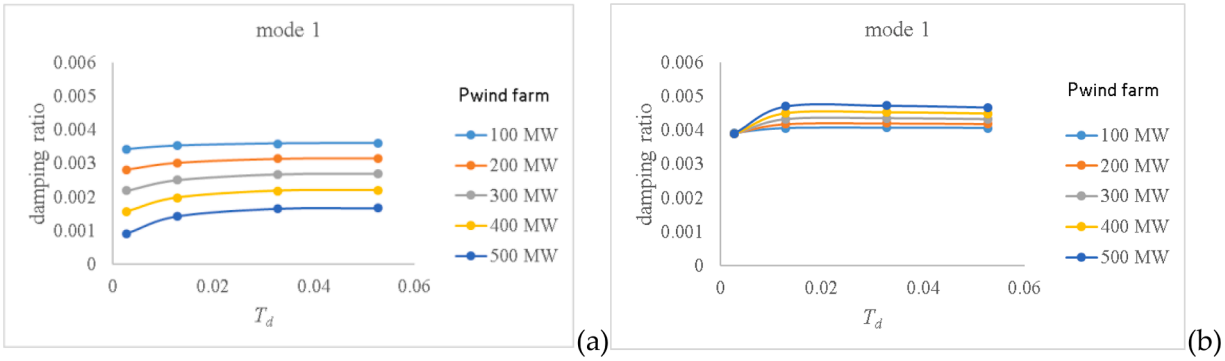


Fig. 21. The influence of generator de-rating operations on the effect of SSTI with integral time constant of current controller for (a) the G1 unit and (b) the G2 unit.

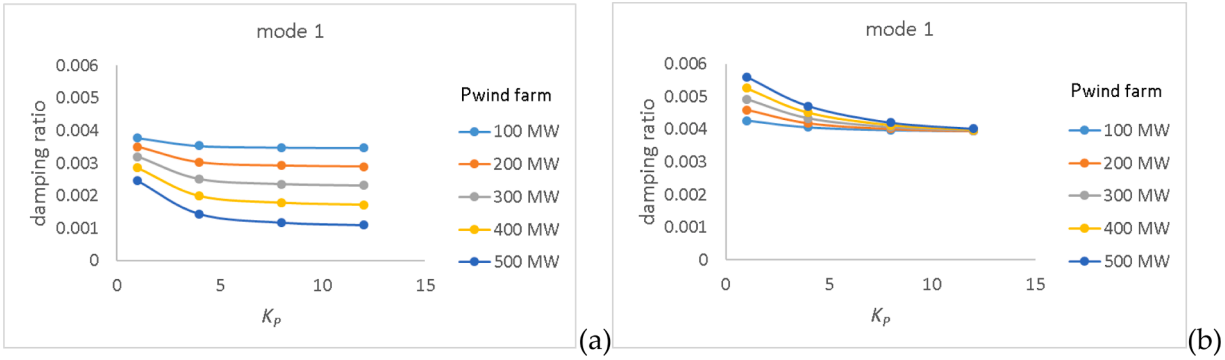


Fig. 22. The influence of generator de-rating operations on the effect of SSTI with proportional gain of power controller for (a) the G1 unit and (b) the G2 unit.

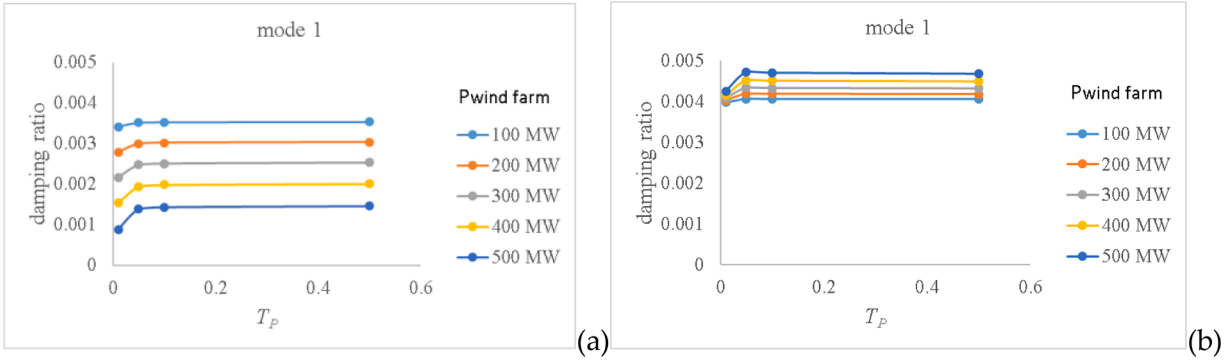


Fig. 23. The influence of generator de-rating operations on the effect of SSTI with integral time constant of power controller for (a) the G1 unit and (b) the G2 unit.

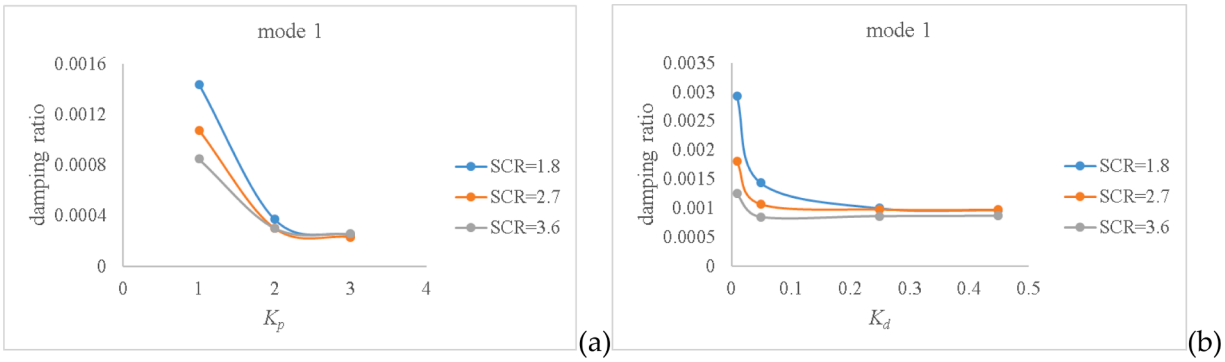


Fig. 24. The SSTI with (a) proportional gain of rectifier controller and (b) proportional gain of DFIG current controller for various SCR.

seen from the figure that the increase in SCR reduces the role of SSTI. Therefore, it can be concluded that the influence of SCR on the SSTI with HVDC controls and on the SSTI with DFIG controls is reducing the SSTI effect. This result is consistent with the general knowledge.

9. Conclusions

In a HVDC transmission system, the transmission power is controlled not to exceed the limitation. The incorporation of a large scale wind farm into the system will lead to that the steam turbine generator unit operates with de-rated power. It is found that the torsional vibrations and stability of torsional responses will be significantly affected by the de-rating operations of the generator unit. Moreover, the generator de-rating operations could significantly influence the effect of SSTI with HVDC controls. After the studies by using the modal analysis and transient simulations, some of the noteworthy results are summarized in the followings.

(1) The torsional mode most sensitive to generator de-rating operations and the torsional mode most sensitive to SSTI are not the same.
(2) The effect of generator de-rating operations on torsional modes is negative. Following the decrease in generation power, the negative damping will be introduced into the torsional modes, leading to some of torsional modes are de-stabilized.
(3) The generator de-rating operation influences the effect of SSTI with HVDC controls in a negative way. It will enlarge the effect of SSTI, and even change the effect from non to negative, from positive to negative or from negative to positive.
(4) The effect of generator de-rating operations and the effect of SSTI with HVDC controls are cumulative. Attention should be paid to the double action of these two effects.
(5) The effect of SSTI with HVDC controls is usually far smaller than the effect of generator de-rating operations. So, the stability of torsional responses is dominated by the effect of generator de-rating operations.
(6) The de-rating operation of a generator unit could exacerbate the effect of SSTI between the HVDC controls and the torsional modes of the neighboring generator unit, and the influence is negative.
(7) The generator de-rating operation would not change the effect of SSTI with DFIG controls, but it does have a cumulative effect.
(8) A large SCR will reduce the SSTI effect.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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